

Inverse Functions of q -Pochhammer Symbols, Gaussian Coefficients, and Related q -Analogues

Local Reversion, Puiseux Branches, Singular Limits,
Root-of-Unity Scaling, and Global Real Inverses

Research report for the ProveIt Fabius/ q -series program

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Abstract

This report develops a systematic inverse-function theory for finite and infinite q -Pochhammer symbols, Gaussian q -binomial coefficients, Jackson's q -gamma function, complex-order shifted factorials, and several related q -analogs. The independent variables being inverted include the base q , the Pochhammer parameter a , the complex order α , the two continuous Gaussian parameters, and the arguments of q -exponentials, q -numbers, Rogers–Szegő polynomials, theta products, and a normalized q -Lambert map.

The guiding point is that “the inverse” is normally not a single function. It is a branched analytic correspondence whose sheets meet above critical values. Ordinary Taylor reversion applies at regular points; Puiseux series appear at multiple zeros and critical points; logarithmic and transseries-like coordinates are required at the singular boundary $q \rightarrow 1$; and algebraic coordinates are natural at $q \rightarrow \pm\infty$ for finite q -products.

Among the concrete results are: explicit inverse series for $(a; q)_n$ and $(a; q)_\infty$ in a and q ; a stable coefficient law near $q = 0$ together with every low-order exception through fifth order; logarithmic inversion at $q = 1$; a complete Puiseux construction at $q = \infty$; exact classification of the exceptional inverse branches at $q = -1$; a root-of-unity radial theorem

$$\log(a; \zeta e^{-t})_\infty = -\frac{\text{Li}_2(a^d)}{d^2 t} + \sum_{r=0}^{d-1} \left(\frac{1}{2} - \frac{r}{d} \right) \text{Log}(1 - a\zeta^r) + \mathcal{O}(t),$$

for primitive d -th roots ζ ; the resulting $a \asymp t^{1/d}$ inverse law; global positive-real inverses of Gaussian polynomials; expansions at $q = 0, 1, \pm\infty$ and roots of unity; exact value and derivative formulas at $q = -1$; inverse theory for continuous Gaussian parameters via q -digamma functions; and $q \rightarrow 0, 1, \infty$ inversion for Γ_q .

A companion Python program performs high-precision numerical and exact symbolic checks. Conjectures are explicitly separated from proved or formally derived statements. The final chapters formulate research programs on monodromy and generic Galois groups, resurgent corrections at the unit circle, certified continuation, coefficient positivity, and formal verification.

Reader's guide and status conventions

The target repository subtree contains two substantial source documents: a frontier volume on exponents and q -series, and a proof-oriented monograph on q -Pochhammer symbols and Gaussian coefficients [1, 2]. They establish a broad *direct* theory: finite and infinite products, logarithmic derivatives, q -gamma representations, q -Newton interpolation, root-of-unity algebra, and $q \rightarrow 1$ asymptotics. They do not organize these facts into a systematic theory of functional inversion. The present report supplies that missing layer and deliberately refers back to the direct theory rather than reproducing its many identity proofs.

Statements marked **proved here** have proofs in this report. Formulae marked **derived here** follow by explicit formal or asymptotic calculation; their analytic domains are stated where needed. Conjectures and open problems are marked **conjectural**. Numerical experiments are consistency checks, never substitutes for proof.

The main recurring notation is

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j),$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}, \quad [x]_q = \frac{1 - q^x}{1 - q}.$$

For singular limits at $q = 1$ we normally write $q = e^{-t}$ or $q = e^t$, according to whether the function is naturally defined inside the unit disk. Branches of Log, fractional powers, and polylogarithms are always part of the data of a complex inverse branch.

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Part I

General architecture of inverse q -theory

Chapter 1

Source audit, scope, and the inverse correspondence

1.1 What has to be inverted

A q -analog usually depends on several parameters, so every direct identity creates several different inverse problems. For the Pochhammer symbol alone one may ask for

$$a = A_n(y; q), \quad q = Q_n(y; a), \quad \alpha = \mathcal{A}(y; a, q)$$

from $y = (a; q)_n$ or $y = (a; q)_\alpha$. The first equation is algebraic of degree n when $n < \infty$; the second is algebraic of degree $n(n-1)/2$; and the third is transcendental but monotone on a useful real domain. No single notation can hide the different branch geometries.

For a Gaussian polynomial, $y = \begin{bmatrix} n \\ k \end{bmatrix}_q$ is an algebraic equation of degree $D = k(n-k)$ in q . If n, k are replaced by continuous parameters α, β through Γ_q , inversion in α is regular and one-sided on the positive real domain, while inversion in β has a symmetry-forced square-root branch at $\beta = \alpha/2$.

1.2 Inverse functions as Riemann surfaces

Let $F(x; \lambda)$ be holomorphic in x and in auxiliary parameters λ . The natural inverse object is

$$\mathcal{X}_F = \{(x, y, \lambda) : F(x; \lambda) = y\} \longrightarrow \{(y, \lambda)\}.$$

Regular sheets are locally graphs $x = X(y; \lambda)$. They meet where $F_x = 0$ or where the direct function itself becomes singular. In polynomial problems the finite branch locus is controlled by

$$\text{Disc}_x(F(x; \lambda) - y) = \text{Res}_x(F(x; \lambda) - y, F_x(x; \lambda)).$$

This viewpoint prevents three common errors: treating a multivalued inverse as single-valued, using Taylor series at a critical point, and confusing a zero of F with a branch point of F^{-1} . A simple zero gives a regular inverse; a multiple zero gives a Puiseux branch.

1.3 A map of the principal regimes

Direct object and variable	Base point	Natural inverse coordinate	Typical geometry
$(a; q)_n$ in a	$a = 0$	$1 - y$	Taylor, unless $[n]_q = 0$
$(a; q)_n$ in a	$a = q^{-r}$	$y^{1/\mu}$	simple or multiple-zero Puisseux
$(a; q)_n$ in q	$q = 0$	$\{y - (1 - a)\}/(a^2 - a)$	stable Taylor reversion
$(a; q)_n$ in q	$q = 1$	$\log\{y/(1 - a)^n\}$	logarithmic Taylor reversion
$(a; q)_n$ in q	$q = \pm\infty$	$y^{-1/N}$	N Puisseux sheets
$(a; q)_\infty$ in q	$q \rightarrow \zeta, q < 1$	$1/(-\log y)$	dilogarithmic singular inversion
$(a; q)_\infty$ in a	$q \rightarrow \zeta$	$t^{1/d}$	root-of-unity Puisseux scaling
$\begin{bmatrix} n \\ k \end{bmatrix}_q$ in q	$q = 0$	$y - 1$	partition-coefficient reversion
$\begin{bmatrix} n \\ k \end{bmatrix}_q$ in q	$q = 1$	$\log\{y/\binom{n}{k}\}$	Bernoulli-cumulant reversion
$\begin{bmatrix} n \\ k \end{bmatrix}_q$ in q	$q = \pm\infty$	$y^{-1/D}$	reciprocal Puisseux sheets
continuous q -binomial in β	$\beta = \alpha/2$	$\sqrt{-\log(y/y_{\max})}$	symmetry-forced double branch
$\Gamma_q(x)$ in x	minimum	$\sqrt{y - y_{\min}}$	two real sheets

Chapter 2

A reusable calculus of reversion and ramification

2.1 Regular local inversion

Theorem 2.1 (Holomorphic inverse and coefficient reversion). *Let*

$$u = f(x) = c_1 x + \frac{c_2}{2!} x^2 + \frac{c_3}{3!} x^3 + \frac{c_4}{4!} x^4 + \cdots, \quad c_1 \neq 0.$$

Then the branch satisfying $x(0) = 0$ is holomorphic and begins

$$\begin{aligned} x = & \frac{u}{c_1} - \frac{c_2}{2c_1^3} u^2 + \frac{3c_2^2 - c_1 c_3}{6c_1^5} u^3 \\ & + \frac{-c_1^2 c_4 + 10c_1 c_2 c_3 - 15c_2^3}{24c_1^7} u^4 + \mathcal{O}(u^5). \end{aligned} \tag{2.1}$$

More generally, if $f(x) = x/\phi(x)$ with $\phi(0) \neq 0$, then

$$[u^m]x(u) = \frac{1}{m} [z^{m-1}] \phi(z)^m.$$

Proof. The analytic assertion is the inverse-function theorem. The coefficient formula is Lagrange–Bürmann inversion. Substituting a formal ansatz $x = b_1 u + b_2 u^2 + \cdots$ and equating coefficients gives (2.1); the calculation takes place in the complete local ring $\mathbb{C}[[u]]$ and therefore also proves formal uniqueness. $\square$

Remark 2.2. For q -products it is usually better to revert a logarithm. If $F(x_0) \neq 0$, set $u = \text{Log}(F(x)/F(x_0))$. Logarithmic derivatives turn products into Lambert sums, and the c_j in (2.1) become cumulants rather than raw derivatives.

2.2 Critical points and Puiseux inversion

Theorem 2.3 (Local Puiseux law). *Suppose*

$$F(x) - y_0 = c_m (x - x_0)^m + c_{m+1} (x - x_0)^{m+1} + \cdots, \quad c_m \neq 0,$$

with $m \geq 2$. Then the inverse correspondence has m local sheets and each has a convergent Puiseux expansion

$$x - x_0 = \omega \left(\frac{y - y_0}{c_m} \right)^{1/m} + \mathcal{O}((y - y_0)^{2/m}), \quad \omega^m = 1.$$

Proof. Write $F(x) - y_0 = (x - x_0)^m G(x)$ with $G(x_0) = c_m \neq 0$. Choose a local holomorphic m -th root $G(x)^{1/m}$ and apply the regular inverse theorem to $(x - x_0)G(x)^{1/m}$. $\square$

The theorem covers multiple Pochhammer zeros, symmetry-forced extrema, and root-of-unity coalescence. Newton polygons provide the analogous construction when two small parameters compete, as in $a \rightarrow 0$ and $q \rightarrow \zeta$.

2.3 Implicit derivatives with respect to every parameter

If $F(x, p) = y$ and $F_x \neq 0$, the inverse branch $x = X(y, p)$ satisfies

$$X_y = \frac{1}{F_x}, \quad X_p = -\frac{F_p}{F_x}, \quad (2.2)$$

$$X_{yy} = -\frac{F_{xx}}{F_x^3}, \quad X_{pp} = -\frac{F_{pp} + 2F_{xp}X_p + F_{xx}X_p^2}{F_x}. \quad (2.3)$$

Mixed higher derivatives are efficiently encoded by multivariate Bell polynomials. Equations (2.2)–(2.3) are the basic engine for expansions in q , a , order, or any auxiliary parameter without re-expanding the whole product.

2.4 Singular asymptotic inversion

Suppose $Y \rightarrow \infty$ and

$$Y = \frac{A}{t} - B + Ct - Et^3 + \mathcal{O}(t^5), \quad A \neq 0.$$

With $\Lambda = Y + B$, direct reversion gives

$$t = \frac{A}{\Lambda} + \frac{A^2 C}{\Lambda^3} + \frac{2A^3 C^2 - A^4 E}{\Lambda^5} + \mathcal{O}(\Lambda^{-7}). \quad (2.4)$$

This simple template will invert the McIntosh expansion of an infinite q -Pochhammer product at $q = 1$, and its root-of-unity analogues.

2.5 Conditioning and rigorous residual bounds

For a real monotone branch, if $|F'(x)| \geq m > 0$ between an approximation $\hat{x}$ and the exact root x_* , then

$$|\hat{x} - x_*| \leq \frac{|F(\hat{x}) - y|}{m}.$$

Thus every truncated inverse series can be turned into a certified enclosure once a derivative lower bound is available. In the complex plane, a Rouché argument on a disk or an interval-Newton/Krawczyk operator gives the analogous certificate. The inverse condition number is

$$\kappa_{\text{abs}} = \frac{1}{|F_x|}, \quad \kappa_{\text{rel}} = \left| \frac{y}{xF_x} \right|,$$

and diverges precisely as a regular sheet approaches the critical locus.

Part II

Finite q -Pochhammer symbols

Chapter 3

Inversion in the Pochhammer parameter a

Throughout this chapter

$$P_n(a, q) := (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad n \geq 1.$$

As a polynomial in a , it has degree n and zeros q^{-r} , $0 \leq r < n$, when $q \neq 0$. Hence a generic level equation $P_n(a, q) = y$ has n complex sheets.

3.1 A global real inverse

Theorem 3.1 (Global a -inverse for $0 < q < 1$). *Fix $0 < q < 1$ and $n \geq 1$. The map*

$$a \mapsto P_n(a, q)$$

is positive and strictly decreasing from $+\infty$ to 0 on $(-\infty, 1)$. Consequently, for every $y > 0$ there is a unique real solution $a < 1$ of $P_n(a, q) = y$.

Proof. For $a < 1$ every factor is positive and

$$\partial_a \log P_n(a, q) = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j} < 0.$$

The limit at $a \rightarrow 1^-$ is 0. As $a \rightarrow -\infty$, the leading term $(-a)^n q^{n(n-1)/2}$ tends to $+\infty$. $\square$

This theorem gives a canonical real sheet. The remaining $n - 1$ sheets are essential for complex continuation and for the $q \rightarrow \infty$ analysis below.

3.2 Reversion at $a = 0$

The finite q -binomial theorem gives

$$P_n(a, q) = \sum_{r=0}^n (-a)^r E_r(n, q), \quad E_r(n, q) = q^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_q. \quad (3.1)$$

Thus $E_1 = [n]_q$. Put $z = 1 - y$.

Proposition 3.2 (Inverse at the unit level). *If $E_1(n, q) \neq 0$, the branch with $a(1) = 0$ is*

$$a = \frac{z}{E_1} + \frac{E_2}{E_1^3} z^2 + \left(\frac{2E_2^2}{E_1^5} - \frac{E_3}{E_1^4} \right) z^3 + \left(\frac{5E_2^3}{E_1^7} - \frac{5E_2E_3}{E_1^6} + \frac{E_4}{E_1^5} \right) z^4 + \mathcal{O}(z^5). \quad (3.2)$$

Proof. Equation (3.1) says $z = E_1 a - E_2 a^2 + E_3 a^3 - E_4 a^4 + \dots$. Apply Theorem 2.1, or equate coefficients directly. $\square$

At $q = 1$, $E_r = \binom{n}{r}$ and (3.2) sums to the exact branch $a = 1 - y^{1/n}$. At a nontrivial root of unity, the hypothesis can fail in a structured way.

Theorem 3.3 (Root-of-unity ramification at $a = 0$). *Let ζ be a primitive d -th root of unity and write $n = md + r$ with $0 \leq r < d$.*

1. *If $r > 0$, then $[n]_\zeta = [r]_\zeta \neq 0$ and the inverse at $a = 0$ is regular.*
2. *If $r = 0$, then*

$$P_n(a, \zeta) = (1 - a^d)^m = 1 - ma^d + \mathcal{O}(a^{2d}),$$

so the inverse has d Puiseux sheets

$$a = \omega \left(\frac{1 - y}{m} \right)^{1/d} + \mathcal{O}((1 - y)^{2/d}), \quad \omega^d = 1.$$

Proof. Group the factors into complete residue blocks modulo d and use $\prod_{j=0}^{d-1} (1 - a\zeta^j) = 1 - a^d$. If a partial block remains, its linear coefficient is $-[r]_\zeta \neq 0$. $\square$

3.3 Inversion at the Pochhammer zeros

Let $a_r = q^{-r}$, $0 \leq r < n$, and assume $q \neq 0$ and that the zeros are distinct. Define

$$R_{n,r}(q) = \prod_{\substack{0 \leq j < n \\ j \neq r}} (1 - q^{j-r}) = (-1)^r q^{-r(r+1)/2} (q; q)_r (q; q)_{n-1-r}.$$

Then

$$\partial_a P_n(a_r, q) = -q^r R_{n,r}(q) = (-1)^{r+1} q^{-r(r-1)/2} (q; q)_r (q; q)_{n-1-r}. \quad (3.3)$$

Proposition 3.4 (Local inverse at a simple zero). *Under the preceding hypotheses, the sheet through a_r is*

$$a = a_r + \frac{y}{P_a(a_r, q)} - \frac{P_{aa}(a_r, q)}{2P_a(a_r, q)^3} y^2 + \mathcal{O}(y^3).$$

At a root of unity several zeros may coalesce.

Proposition 3.5 (Multiplicity at a root of unity). *Let ζ be primitive of order d , let $n = md + r$, $0 \leq r < d$, and let $a_0 = \zeta^{-s}$, $0 \leq s < d$. As a function of a , the zero of $P_n(a, \zeta)$ at a_0 has multiplicity*

$$\mu_s = m + \mathbf{1}_{\{0 \leq s < r\}}.$$

Consequently, the inverse at $y = 0$ has μ_s Puiseux sheets.

Proof. A factor vanishes exactly when its index is congruent to s modulo d . Among $0, \dots, n-1$, that residue occurs m times and once more precisely when it belongs to the partial block $0, \dots, r-1$. $\square$

3.4 Exact inverses at $q = 0, 1, -1$

Three bases collapse the a -equation to elementary algebra.

At $q = 0$,

$$P_n(a, 0) = 1 - a, \quad a = 1 - y.$$

At $q = 1$,

$$P_n(a, 1) = (1 - a)^n, \quad a = 1 - \omega y^{1/n}, \quad \omega^n = 1.$$

At $q = -1$, put $u = \lceil n/2 \rceil$, $v = \lfloor n/2 \rfloor$. Then

$$P_n(a, -1) = (1 - a)^u (1 + a)^v. \quad (3.4)$$

If $n = 2m$, this becomes $(1 - a^2)^m$, and the inverse is exactly

$$a = \pm \sqrt{1 - y^{1/m}},$$

with all choices of the m -th root included in the complex correspondence. The double branch at $a = 0, y = 1$ is forced by evenness. If $n = 2m + 1$, then $P_n = (1 - a)(1 - a^2)^m$ and the sheet at $a = 0$ is regular because $P_a(0, -1) = -1$.

3.5 The a -inverse for q near 0 and 1

Fix y and regard $a = a(q; y)$. For the branch satisfying $a(0) = 1 - y$, formal implicit reversion gives, for $n \geq 5$ through the displayed order,

$$\begin{aligned} a &= 1 - y + y(y - 1)q - 2y(y - 1)^2 q^2 \\ &\quad + y(y - 1)^2(5y - 3)q^3 - y(y - 1)^2(2y - 1)(7y - 5)q^4 + \mathcal{O}(q^5). \end{aligned} \quad (3.5)$$

The coefficient of q^M stabilizes once $n > M$ because the omitted factors begin at q^n .

For $q = e^t$ near 1, choose an n -th root $r = y^{1/n}$ and the branch $a(0) = 1 - r$. Implicit differentiation yields

$$\begin{aligned} a(t) &= 1 - r - \frac{(n - 1)(1 - r)}{2} t \\ &\quad + \frac{(n - 1)(1 - r)(3nr - n - 3r - 1)}{24r} t^2 + \mathcal{O}(t^3). \end{aligned} \quad (3.6)$$

The different choices of r parametrize the n complex sheets.

3.6 All a -branches as $q \rightarrow \infty$

The polynomial equation $P_n(a, q) = y$ has n asymptotic a -branches. This is more informative than following only the global real branch.

Theorem 3.6 (a -branch atlas at large base). *Let $y \neq 0$ be fixed and $q \rightarrow \infty$ in a sector. For $0 \leq r \leq n - 2$, put*

$$M_r = \frac{(n - 1 - r)(n - r)}{2}.$$

There is a branch near the moving zero q^{-r} with

$$a_r(q) = q^{-r} \left(1 + (-1)^{n-r} y q^{-M_r} + \mathcal{O}(q^{-M_r-1}) + \mathcal{O}(q^{-2M_r}) \right). \quad (3.7)$$

The remaining branch has the full expansion

$$\begin{aligned} a_{n-1}(q) = q^{-(n-1)} & (1 - y + y(y-1)q^{-1} - 2y(y-1)^2q^{-2} \\ & + y(y-1)^2(5y-3)q^{-3} + \mathcal{O}(q^{-4})). \end{aligned} \quad (3.8)$$

The same formal Laurent series describe $q \rightarrow -\infty$, with the signs already encoded by powers of q^{-1} .

Proof. For $r \leq n-2$, set $a = q^{-r}(1 + \varepsilon)$. The vanishing factor is $1 - aq^r = -\varepsilon$, factors with $j < r$ tend to 1, and those with $j > r$ have product $(-1)^{n-1-r} q^{M_r}(1 + o(1))$. Solving the leading equation gives (3.7). For the final branch write $a = q^{-(n-1)}c$ and $s = q^{-1}$. Then

$$P_n(a, q) = (c; s)_n.$$

The branch $c(0) = 1 - y$ is exactly the small- s inverse (3.5), proving (3.8). $\square$

Chapter 4

Inversion in the base q

Let

$$N = \frac{n(n-1)}{2}.$$

For fixed a , $P_n(a, q)$ is a polynomial of degree N in q , except in degenerate cases. Hence a generic base inverse has N complex branches.

4.1 A canonical real branch on $0 \leq q \leq 1$

Theorem 4.1 (Global finite base inverse). *Let $n \geq 2$.*

1. *If $0 < a < 1$, then $q \mapsto P_n(a, q)$ is strictly decreasing on $[0, 1]$ from $1 - a$ to $(1 - a)^n$.*
2. *If $a < 0$, it is strictly increasing on $[0, 1]$ from $1 - a$ to $(1 - a)^n$.*

Thus every level strictly between the endpoint values has one and only one real inverse $q \in (0, 1)$.

Proof. For $0 < q < 1$,

$$\partial_q \log P_n(a, q) = -a \sum_{j=1}^{n-1} \frac{j q^{j-1}}{1 - a q^j}.$$

All denominators are positive. The sign is therefore $-\operatorname{sgn} a$. □

The cases $a = 0$, $a = 1$, $n = 0$, and $n = 1$ are constant in q and have no local base inverse.

4.2 The stable inverse series at $q = 0$

For $a \notin \{0, 1\}$ define the normalized level coordinate

$$x = \frac{y - (1 - a)}{a^2 - a}. \tag{4.1}$$

Then $x = q + \mathcal{O}(q^2)$.

Theorem 4.2 (Stable small-base reversion). *For the infinite formal product, and for the finite product whenever $n > 5$,*

$$\begin{aligned} q = x - x^2 + (a + 1)x^3 - (4a + 1)x^4 \\ + (3a^2 + 11a + 1)x^5 + \mathcal{O}(x^6). \end{aligned} \tag{4.2}$$

More generally, the coefficient through x^M agrees with the infinite-product coefficient whenever $n > M$.

Proof. The quotient of $(a; q)_\infty$ by $(a; q)_n$ is $(aq^n; q)_\infty = 1 + \mathcal{O}(q^n)$. Therefore the direct q -series agree through q^{n-1} , and formal inversion preserves agreement through the same order. Direct reversion gives (4.2). $\square$

The low-degree exceptions are important enough to record:

$$\begin{aligned} n = 2 : \quad & q = x, \\ n = 3 : \quad & q = x - x^2 + (a + 2)x^3 - 5(a + 1)x^4 + (3a^2 + 21a + 14)x^5 + \mathcal{O}(x^6), \\ n = 4 : \quad & q = x - x^2 + (a + 1)x^3 - 4ax^4 + (3a^2 + 10a - 4)x^5 + \mathcal{O}(x^6), \\ n = 5 : \quad & q = x - x^2 + (a + 1)x^3 - (4a + 1)x^4 + (3a^2 + 11a + 2)x^5 + \mathcal{O}(x^6). \end{aligned}$$

These identities were also checked exactly in the companion SymPy script.

4.3 Logarithmic inversion at $q = 1$

Set $q = e^t$ and

$$u = \text{Log} \frac{P_n(a, e^t)}{(1 - a)^n}.$$

Let $S_m(n) = \sum_{j=0}^{n-1} j^m$. Expanding the logarithm of every factor gives the exact jet

$$u = \sum_{m \geq 1} \frac{c_m}{m!} t^m, \quad c_m = -S_m(n) \text{Li}_{1-m}(a). \quad (4.3)$$

Thus the inverse $t = t(u)$ is obtained from (2.1). Explicitly,

$$t = \frac{u}{c_1} - \frac{c_2}{2c_1^3} u^2 + \frac{3c_2^2 - c_1 c_3}{6c_1^5} u^3 + \mathcal{O}(u^4), \quad (4.4)$$

$$q = \exp(t). \quad (4.5)$$

The first direct derivative is

$$\partial_q P_n(a, 1) = -aS_1(n)(1 - a)^{n-1},$$

so, in the raw level coordinate,

$$q - 1 = -\frac{y - (1 - a)^n}{aS_1(n)(1 - a)^{n-1}} + \mathcal{O}((y - (1 - a)^n)^2). \quad (4.6)$$

The inverse is regular unless $a = 0$, $a = 1$, or $n \leq 1$.

4.4 The complete Puiseux construction at $q = \infty$

Factor

$$P_n(a, q) = C(a)q^N H(q^{-1}), \quad C(a) = (1 - a)(-a)^{n-1}, \quad H(s) = \prod_{j=1}^{n-1} (1 - a^{-1}s^j). \quad (4.7)$$

Assume $a \notin \{0, 1\}$ and $y \rightarrow \infty$. Choose an N -th root $x = (C/y)^{1/N}$ and solve

$$s = xH(s)^{1/N}, \quad q = s^{-1}. \quad (4.8)$$

Lagrange inversion gives the all-orders coefficient formula

$$[x^m]s(x) = \frac{1}{m}[u^{m-1}]H(u)^{m/N}. \quad (4.9)$$

Every choice of x gives one sheet. For $n \geq 3$ the first terms are

$$q = x^{-1} + \frac{1}{aN} + \left(\frac{1}{aN} + \frac{N-1}{2N^2a^2} \right) x + \mathcal{O}(x^2). \quad (4.10)$$

For $n = 2$, this reduces to the exact linear inverse $q = x^{-1} + a^{-1}$. The negative-real asymptotic $q \rightarrow -\infty$ is not a separate formula: it is the sheet for which the leading root x^{-1} is negative real. The parity of N and the sign of $C(a)$ determine which real levels reach that sheet.

4.5 Zeros in the q -plane

For fixed $a \neq 0$, every equation

$$q^r = a^{-1}, \quad 1 \leq r < n,$$

produces zeros of P_n as a function of q . If q_0 makes only the r -th factor vanish, then

$$P_q(a, q_0) = -arq_0^{r-1} \prod_{\substack{0 \leq j < n \\ j \neq r}} (1 - aq_0^j), \quad (4.11)$$

and the inverse at $y = 0$ is regular. If several indices satisfy $aq_0^j = 1$, their number is the zero multiplicity and [Theorem 2.3](#) applies. Such collisions occur precisely on arithmetic strata involving roots of unity.

4.6 The exceptional geometry at $q = -1$

For $a \neq \pm 1$, the logarithmic base derivative at $q = -1$ is

$$\frac{P_q(a, -1)}{P_n(a, -1)} = a \left(\frac{S_{\text{even}}}{1-a} - \frac{S_{\text{odd}}}{1+a} \right), \quad (4.12)$$

where the sums run over $0 \leq j < n$. Evaluation gives the following complete classification apart from constant degeneracies.

Theorem 4.3 (Critical Pochhammer parameters at $q = -1$). *1. If $n = 2m$, then*

$$\frac{P_q}{P} = \frac{am((2m-1)a-1)}{1-a^2}.$$

For $m \geq 2$, the nontrivial critical parameter is $a_c = (2m-1)^{-1}$ and

$$\left. \frac{P_{qq}}{P} \right|_{(a_c, -1)} = -\frac{(m+1)(2m-1)^2}{12m(m-1)} \neq 0.$$

The inverse therefore has a square-root branch at $y_c = P_n(a_c, -1)$.

2. If $n = 2m + 1$, then

$$\frac{P_q}{P} = \frac{am(1 + (2m + 1)a)}{1 - a^2}.$$

For $m \geq 2$, the critical parameter is $a_c = -(2m + 1)^{-1}$ and

$$\left. \frac{P_{qq}}{P} \right|_{(a_c, -1)} = -\frac{(m - 1)(2m + 1)^2}{12m(m + 1)} \neq 0.$$

Again the inverse is square-root ramified.

3. The exceptional case $n = 3$, $a = -1/3$ is cubic rather than quadratic:

$$P_3(-1/3, q) = \frac{32}{27} + \frac{4}{27}(q + 1)^3.$$

Hence

$$q = -1 + \left(\frac{27}{4} \left(y - \frac{32}{27} \right) \right)^{1/3}$$

with three local branches.

Proof. Equation (4.12) follows by differentiating the product. The even and odd arithmetic sums give the displayed first derivatives. A second differentiation and elementary power sums give the two ratios for P_{qq}/P . The cubic identity in the last case is an exact expansion. $\square$

4.7 General roots of unity

Let $q = \zeta$ be primitive of order d . The direct product value is

$$P_n(a, \zeta) = (1 - a^d)^m (a; \zeta)_r, \quad n = md + r, \quad 0 \leq r < d.$$

The derivative criterion

$$-a \sum_{j=1}^{n-1} \frac{j\zeta^{j-1}}{1 - a\zeta^j} \neq 0$$

ensures an ordinary local q -inverse. Its vanishing defines an explicit algebraic critical locus in a . When a factor also vanishes, one first extracts all zero factors and applies the Newton polygon to the residual two-variable germ. This produces a finite, algorithmic classification of root-of-unity branches for every fixed n .

Part III

Infinite products and complex order

Chapter 5

The infinite q -Pochhammer inverse

Assume initially $|q| < 1$. The product

$$P_\infty(a, q) := (a; q)_\infty$$

is entire in a , while its dependence on q is naturally tied to the unit disk. The Lambert-series identity

$$\text{Log}(a; q)_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - q^m)} \quad (5.1)$$

holds absolutely for $|a| < 1$, $|q| < 1$ and extends by analytic continuation in a away from chosen logarithmic cuts. It is the central tool for inverse asymptotics.

5.1 Global real inverses in a and q

Theorem 5.1 (Global real sheets for the infinite product). *Fix $0 < q < 1$. The function $a \mapsto (a; q)_\infty$ is positive and strictly decreasing from $+\infty$ to 0 on $(-\infty, 1)$, so it has a unique real a -inverse for every $y > 0$.*

Fix instead $a \in (-\infty, 1) \setminus \{0\}$. On $0 \leq q < 1$ the function is strictly decreasing when $0 < a < 1$ and strictly increasing when $a < 0$. Its ranges are

$$0 < a < 1 : \quad (0, 1 - a], \quad a < 0 : \quad [1 - a, \infty).$$

Hence it has a unique real base inverse on the corresponding range.

Proof. The a -derivative is

$$\partial_a \text{Log}(a; q)_\infty = - \sum_{j \geq 0} \frac{q^j}{1 - aq^j} < 0$$

for $a < 1$. The large negative- a limit follows, for example, by retaining an increasing number of factors with $|a|q^j \geq 1$. The base derivative has the same sign calculation as in [Theorem 4.1](#). Finally, (5.1) and the $q \rightarrow 1$ asymptotic below show convergence to 0 for $0 < a < 1$ and divergence to $+\infty$ for $a < 0$. $\square$

5.2 The a -inverse at $a = 0$

Euler's expansion gives

$$(a; q)_\infty = \sum_{r \geq 0} \frac{(-1)^r q^{r(r-1)/2}}{(q; q)_r} a^r.$$

Substitution into (3.2) yields an explicit branch. With $z = 1 - y$,

$$\begin{aligned} a &= (1 - q)z + \frac{q(1 - q)}{1 + q} z^2 \\ &\quad + \frac{q^2(1 - q)(q^2 + q + 2)}{(1 + q)^2(1 + q + q^2)} z^3 + \mathcal{O}(z^4). \end{aligned} \quad (5.2)$$

This expansion is valid away from roots of the displayed cyclotomic denominators, and elsewhere after the appropriate Puiseux rescaling.

5.3 The a -inverse at the first zero $a = 1$

Let

$$C(q) = (q; q)_\infty, \quad S_r(q) = \sum_{j \geq 1} \left(\frac{q^j}{1 - q^j} \right)^r, \quad w = \frac{y}{C(q)}.$$

Writing $z = 1 - a$ gives

$$(a; q)_\infty = C(q)z \exp \left(S_1 z - \frac{S_2}{2} z^2 + \frac{S_3}{3} z^3 - \dots \right).$$

Therefore

$$1 - a = w - S_1 w^2 + \frac{3S_1^2 + S_2}{2} w^3 + \mathcal{O}(w^4). \quad (5.3)$$

At the general zero $a_r = q^{-r}$, $r \geq 0$, one has

$$\partial_a (a; q)_\infty \big|_{a=q^{-r}} = (-1)^{r+1} q^{-r(r-1)/2} (q; q)_r (q; q)_\infty, \quad (5.4)$$

so every zero is simple when q is not a root of unity and the local inverse is ordinary Taylor.

5.4 The base inverse at $q = 0$

The stable finite formula (4.2) is exactly the infinite formula:

$$x = \frac{y - (1 - a)}{a^2 - a},$$

$$q = x - x^2 + (a + 1)x^3 - (4a + 1)x^4 + (3a^2 + 11a + 1)x^5 + \mathcal{O}(x^6). \quad (5.5)$$

Conversely, fixing y and inverting in a gives the all-product version of (3.5):

$$\begin{aligned} a &= 1 - y + y(y - 1)q - 2y(y - 1)^2 q^2 \\ &\quad + y(y - 1)^2 (5y - 3)q^3 - y(y - 1)^2 (2y - 1)(7y - 5)q^4 + \mathcal{O}(q^5). \end{aligned} \quad (5.6)$$

5.5 Singular base inversion at $q \rightarrow 1^-$

Put $q = e^{-t}$, $t \downarrow 0$, and fix a in a compact subset of a domain where the chosen branches are analytic. The McIntosh expansion [8, 7] is

$$\begin{aligned} \text{Log}(a; e^{-t})_\infty = & -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \text{Log}(1-a) \\ & - \sum_{r=1}^{M-1} \frac{B_{2r}}{(2r)!} t^{2r-1} \text{Li}_{2-2r}(a) + \mathcal{O}(t^{2M-1}). \end{aligned} \quad (5.7)$$

For $0 < a < 1$, let

$$\begin{aligned} Y = -\log y, \quad A = \text{Li}_2(a), \quad B = \frac{1}{2} \log(1-a), \\ C = \frac{\text{Li}_0(a)}{12} = \frac{a}{12(1-a)}, \quad E = \frac{\text{Li}_{-2}(a)}{720} = \frac{a(1+a)}{720(1-a)^3}, \quad \Lambda = Y + B. \end{aligned}$$

Then (2.4) gives the base inverse

$$t = \frac{A}{\Lambda} + \frac{A^2 C}{\Lambda^3} + \frac{2A^3 C^2 - A^4 E}{\Lambda^5} + \mathcal{O}(\Lambda^{-7}), \quad q = e^{-t}. \quad (5.8)$$

In particular,

$$q = 1 - \frac{A}{\Lambda} + \frac{A^2}{2\Lambda^2} - \frac{A^3/6 + A^2 C}{\Lambda^3} + \mathcal{O}(\Lambda^{-4}). \quad (5.9)$$

The numerical table in the appendix shows the expected gain of two inverse powers of Λ at each correction.

Remark 5.2 (Complex sectors and exponentially small terms). Formula (5.7) is a Poincaré expansion. A complete global inverse requires a branch of $\text{Li}_2(a)$ and $\text{Log}(1-a)$ and, near Stokes rays, exponentially small modular or resurgence corrections. These effects are invisible to algebraic powers of $1/\Lambda$ but can move the inverse by an amount comparable with the least term of the expansion.

5.6 Fixed level, a -inversion as $q \rightarrow 1$

The previous regime holds a fixed and sends y to zero. A different and important inverse problem fixes $0 < y < 1$ and lets $a = a(t)$ tend to zero. Put $L = -\log y$. Substituting a power series into the Lambert sum gives

$$\begin{aligned} a(t) = Lt - \frac{L(L+2)}{4} t^2 + \frac{L(L^2+9L+12)}{72} t^3 \\ - \frac{L(L^3+4L^2+12L+24)}{576} t^4 + \mathcal{O}(t^5). \end{aligned} \quad (5.10)$$

This is a regular series in t , not in $1/\log y$. It describes the moving level curves that terminate at the corner $(a, q) = (0, 1)$.

Derivation. From (5.1),

$$L = \sum_{m \geq 1} \frac{a^m}{m(1 - e^{-mt})}.$$

Use $1/(1 - e^{-x}) = x^{-1} + \frac{1}{2} + x/12 - x^3/720 + \dots$, insert $a = c_1 t + c_2 t^2 + \dots$, and solve successively. The first four equations give (5.10). $\square$

5.7 Radial asymptotics at every root of unity

The singular point $q = 1$ is one member of a family. Let ζ be primitive of order d and approach it radially by

$$q = \zeta e^{-t}, \quad t \downarrow 0.$$

Group factors by their residue class modulo d :

$$(a; \zeta e^{-t})_\infty = \prod_{r=0}^{d-1} (a\zeta^r e^{-rt}; e^{-dt})_\infty. \quad (5.11)$$

Theorem 5.3 (Root-of-unity dilogarithm law). *Let $|a| < 1$ and fix compatible logarithm branches. As $t \rightarrow 0$ in a sector,*

$$\text{Log}(a; \zeta e^{-t})_\infty = -\frac{\text{Li}_2(a^d)}{d^2 t} + C_{d,\zeta}(a) + \mathcal{O}(t), \quad (5.12)$$

$$C_{d,\zeta}(a) = \sum_{r=0}^{d-1} \left(\frac{1}{2} - \frac{r}{d} \right) \text{Log}(1 - a\zeta^r). \quad (5.13)$$

Proof. Apply (5.7) with base e^{-dt} to every factor in (5.11). The leading term is

$$-\frac{1}{dt} \sum_{r=0}^{d-1} \text{Li}_2(a\zeta^r).$$

The root-of-unity filter gives

$$\sum_{r=0}^{d-1} \text{Li}_2(a\zeta^r) = \frac{1}{d} \text{Li}_2(a^d).$$

The parameter $a\zeta^r e^{-rt}$ contributes $-(r/d) \text{Log}(1 - a\zeta^r)$ from the first-order expansion of its dilogarithm, while McIntosh contributes $\frac{1}{2} \text{Log}(1 - a\zeta^r)$. Summing gives (5.13). $\square$

Corollary 5.4 (Base inverse near a root of unity). *Let $y = (a; q)_\infty$ tend to zero along a branch for which $A_d = \text{Li}_2(a^d)/d^2 \neq 0$. Put $Y = -\text{Log } y$ and $\Lambda = Y + C_{d,\zeta}(a)$. Then*

$$t = \frac{A_d}{\Lambda} + \mathcal{O}(\Lambda^{-3}), \quad q = \zeta \exp \left(-\frac{A_d}{\Lambda} + \mathcal{O}(\Lambda^{-3}) \right).$$

Higher terms follow by carrying (5.12) to higher order and applying (2.4).

Corollary 5.5 (Universal fixed-level Puiseux scale). *Fix $y = e^{-L}$ with $L \neq 0$. On branches tending to $a = 0$,*

$$a^d \sim d^2 L t, \quad a \sim \omega (d^2 L t)^{1/d}, \quad \omega^d = 1. \quad (5.14)$$

Thus radial inversion in a has exponent $1/d$ at a primitive d -th root of unity.

Proof. For small a , $\text{Li}_2(a^d) = a^d + \mathcal{O}(a^{2d})$. Insert this into (5.12) while keeping L fixed. $\square$

The exponent $1/d$ is the analytic counterpart of the exact finite identity $(a; \zeta)_{md} = (1 - a^d)^m$.

5.8 The radial point $q = -1$ in detail

Take $d = 2$, $\zeta = -1$. The exact factorization is

$$(a; -e^{-t})_{\infty} = (a; e^{-2t})_{\infty} (-ae^{-t}; e^{-2t})_{\infty}.$$

For $0 < a < 1$, expansion to the first correction gives

$$\log(a; -e^{-t})_{\infty} = -\frac{\text{Li}_2(a^2)}{4t} + \frac{1}{2} \log(1 - a) - \frac{a(a+3)}{12(1-a^2)}t + \mathcal{O}(t^3). \quad (5.15)$$

There is no t^2 term. If

$$A_- = \frac{\text{Li}_2(a^2)}{4}, \quad B_- = \frac{1}{2} \log(1 - a), \quad C_- = \frac{a(a+3)}{12(1-a^2)},$$

then

$$t = \frac{A_-}{Y + B_-} + \frac{A_-^2 C_-}{(Y + B_-)^3} + \mathcal{O}(Y^{-5}), \quad q = -e^{-t}. \quad (5.16)$$

For the complementary fixed-level inverse, $a = \mathcal{O}(t^{1/2})$. Solving the Lambert series in powers of $s = t^{1/2}$ gives the more precise positive branch

$$a = 2\sqrt{Lt} - t + \frac{\frac{1}{4} - L - L^2}{\sqrt{L}} t^{3/2} + \frac{4L + 3}{6} t^2 + \mathcal{O}(t^{5/2}). \quad (5.17)$$

The negative branch and the complex branches are obtained by the appropriate choice of the leading square root, followed by coefficient recursion.

5.9 Why $q = \pm\infty$ is different for the infinite product

The product $(a; q)_{\infty}$ does not converge when $|q| > 1$. Therefore a statement about its inverse at $q = \pm\infty$ is meaningless until a specific regularization or reciprocal-base continuation is chosen. Finite products have the canonical algebraic theory of [Equation \(4.10\)](#); complex orders can be continued using reciprocal-base definitions; but two different regularizations of the infinite product can differ by exponential factors and produce different inverse asymptotics. This report consequently treats $q = \pm\infty$ canonically for finite products and for functions such as Γ_q that possess a standard reciprocal-base definition.

Chapter 6

Inversion in the complex order and truncation index

For $|q| < 1$, define the complex-order shifted factorial by

$$(a; q)_\alpha = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}, \quad (6.1)$$

with a chosen branch of $q^\alpha = e^{\alpha \operatorname{Log} q}$.

6.1 Global real order inverse

Theorem 6.1 (Monotone order branch). *Let $0 < a, q < 1$. The map*

$$\alpha \longmapsto (a; q)_\alpha$$

is strictly decreasing on $[0, \infty)$ from 1 to $(a; q)_\infty$. Consequently every $y \in ((a; q)_\infty, 1]$ has a unique real inverse order.

Proof. Differentiating (6.1) gives

$$\partial_\alpha \operatorname{Log}(a; q)_\alpha = \operatorname{Log} q \sum_{j \geq 0} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}} < 0. \quad (6.2)$$

The endpoints follow directly from the definition. □

6.2 Series at order $\alpha = 0$

Put $u = \operatorname{Log} y$ and

$$c_m = (\operatorname{Log} q)^m \sum_{j \geq 0} \operatorname{Li}_{1-m}(aq^j), \quad m \geq 1.$$

Then

$$\operatorname{Log}(a; q)_\alpha = \sum_{m \geq 1} \frac{c_m}{m!} \alpha^m,$$

and hence

$$\alpha = \frac{u}{c_1} - \frac{c_2}{2c_1^3} u^2 + \frac{3c_2^2 - c_1 c_3}{6c_1^5} u^3 + \mathcal{O}(u^4). \quad (6.3)$$

The Lambert sums converge rapidly for fixed $q < 1$ and give a practical local inverse without numerical differencing.

6.3 Large order and inversion of the tail

Let

$$P_\infty = (a; q)_\infty, \quad w = \frac{y}{P_\infty} - 1, \quad z = aq^\alpha.$$

As $\alpha \rightarrow \infty$, Euler's reciprocal expansion gives

$$1 + w = \frac{1}{(z; q)_\infty} = 1 + \frac{z}{1 - q} + \frac{z^2}{(1 - q)(1 - q^2)} + \cdots.$$

Reversion yields

$$\begin{aligned} z &= (1 - q)w - \frac{1 - q}{1 + q}w^2 \\ &\quad + \frac{(1 - q)(2q^2 + q + 1)}{(1 + q)^2(1 + q + q^2)}w^3 + \mathcal{O}(w^4). \end{aligned} \tag{6.4}$$

Therefore

$$\begin{aligned} \alpha &= \frac{1}{\text{Log } q} \left[\text{Log } \frac{(1 - q)w}{a} - \frac{w}{1 + q} \right. \\ &\quad \left. + \frac{3q^2 + q + 1}{2(1 + q)^2(1 + q + q^2)}w^2 + \mathcal{O}(w^3) \right]. \end{aligned} \tag{6.5}$$

This logarithmic divergence quantifies how many factors are needed before a finite product is within a prescribed relative distance of the infinite one.

6.4 Recovering a discrete truncation index

For integer n the sequence $(a; q)_n$ decreases to P_∞ . Define

$$n_-(y) = \max\{n : (a; q)_n \geq y\}, \quad P_\infty < y < 1.$$

The continuous inverse (6.5) gives an asymptotically sharp estimate, and the exact discrete index is one of the two neighboring integers. A certified algorithm evaluates those two finite products, avoiding any ambiguity caused by rounding a slowly varying logarithm.

6.5 Joint inversion and level-curve differential equations

For $F(a, q, \alpha) = (a; q)_\alpha$, a level surface $F = y$ satisfies

$$F_a da + F_q dq + F_\alpha d\alpha = 0.$$

Thus, for example,

$$\left. \frac{d\alpha}{dq} \right|_{a,y} = -\frac{F_q}{F_\alpha}, \quad \left. \frac{da}{dq} \right|_{\alpha,y} = -\frac{F_q}{F_a}.$$

The logarithmic derivative representations make these ratios explicit Lambert series. They are useful both analytically, for proving monotonicity, and numerically, as tangent predictors in continuation algorithms.

Part IV

Gaussian coefficients and q -gamma inverses

Chapter 7

The base inverse of a Gaussian polynomial

Let

$$B_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad 0 < k < n, \quad D = k(n-k).$$

It is a monic reciprocal polynomial of degree D with nonnegative integer coefficients:

$$B_{n,k}(q) = q^D B_{n,k}(q^{-1}). \quad (7.1)$$

Its coefficient of q^m counts partitions of m fitting inside a $k \times (n-k)$ rectangle.

7.1 The unique positive inverse

Theorem 7.1 (Global positive Gaussian inverse). *The map $q \mapsto B_{n,k}(q)$ is strictly increasing from 1 to $+\infty$ on $[0, \infty)$. Therefore every $y \geq 1$ has a unique inverse $Q_{n,k}(y) \geq 0$. For $y \geq \binom{n}{k}$,*

$$\left(\frac{y}{\binom{n}{k}} \right)^{1/D} \leq Q_{n,k}(y) \leq y^{1/D}. \quad (7.2)$$

Proof. Every nonconstant coefficient is nonnegative and the coefficient of q is 1, so the derivative is positive for $q \geq 0$. For $q \geq 1$, reciprocity and positivity give $q^D \leq B_{n,k}(q) \leq B_{n,k}(1)q^D = \binom{n}{k}q^D$. Invert these inequalities. $\square$

This theorem isolates a canonical sheet among the D complex branches. It is also the best starting point for certified numerical inversion because bisection never loses the branch.

7.2 Partition-controlled reversion at $q = 0$

Write

$$B_{n,k}(q) = 1 + \sum_{m \geq 1} p_{k,n-k}(m) q^m,$$

where $p_{r,s}(m)$ counts partitions of m in an $r \times s$ rectangle. Put $s = y - 1$ and abbreviate

$$p_2 = \mathbf{1}_{\{k \geq 2\}} + \mathbf{1}_{\{n-k \geq 2\}},$$

$$p_3 = \mathbf{1}_{\{n-k \geq 3\}} + \mathbf{1}_{\{k \geq 2, n-k \geq 2\}} + \mathbf{1}_{\{k \geq 3\}}.$$

Then

$$q = s - p_2 s^2 + (2p_2^2 - p_3) s^3 + \mathcal{O}(s^4). \quad (7.3)$$

All higher coefficients are universal polynomials in the rectangle partition numbers. Lagrange–Bürmann gives the compact all-orders formula

$$[s^m]Q_{n,k}(1+s) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{B_{n,k}(z) - 1} \right)^m. \quad (7.4)$$

7.3 Bernoulli-cumulant inversion at $q = 1$

Set $q = e^t$, $C_0 = \binom{n}{k}$, and $u = \log(B_{n,k}(e^t)/C_0)$. For

$$A_m(n, k) = \sum_{j=1}^k ((n-k+j)^m - j^m),$$

the exact formal expansion is

$$u = \frac{D}{2}t + \sum_{r \geq 1} \frac{B_{2r} A_{2r}(n, k)}{2r(2r)!} t^{2r}. \quad (7.5)$$

In particular,

$$A_2 = D(n+1), \quad A_4 = D(n+1)(n(n+1) - D).$$

Reversion gives

$$t = \frac{2u}{D} - \frac{n+1}{3D^2} u^2 + \frac{(n+1)^2}{9D^3} u^3 + \frac{(n+1)\{6[n(n+1) - D] - 25(n+1)^2\}}{540D^4} u^4 + \mathcal{O}(u^5), \quad (7.6)$$

$$q = e^t. \quad (7.7)$$

Proof of the cumulant formula. For every positive integer m ,

$$\log \frac{e^{mt} - 1}{m(e^t - 1)} = \frac{m-1}{2}t + \sum_{r \geq 1} \frac{B_{2r}(m^{2r} - 1)}{2r(2r)!} t^{2r}.$$

Apply this identity to the product formula for $B_{n,k}$ and sum over $j = 1, \dots, k$. The inverse series follows from ordinary reversion. $\square$

The absence of odd cumulants beyond the linear term is a reciprocity effect. Indeed (7.1) implies that $e^{-Dt/2} B_{n,k}(e^t)$ is an even function of t .

7.4 Puiseux inversion at $q = \pm\infty$

Let $R = y^{1/D}$ on a chosen branch. By reciprocity,

$$B_{n,k}(q) = q^D \left(1 + q^{-1} + p_2 q^{-2} + \mathcal{O}(q^{-3}) \right).$$

Therefore

$$q = R - \frac{1}{D} + \left(\frac{D-1}{2D^2} - \frac{p_2}{D} \right) R^{-1} + \mathcal{O}(R^{-2}). \quad (7.8)$$

The D choices of R are the D inverse sheets at infinity. A real $q \rightarrow -\infty$ branch exists for those real levels whose sign agrees with q^D ; in particular, it reaches positive large y when D is even and negative large y when D is odd.

The full expansion is algorithmic. Substitute $q = R + c_0 + c_1 R^{-1} + \dots$ into $q^D B_{n,k}(q^{-1}) = R^D$ and solve recursively. Only the first m partition coefficients are required for the inverse through R^{1-m} .

7.5 Exact local data at $q = -1$

The point $q = -1$ is especially clean.

Theorem 7.2 (Value and derivative at -1). *Write $n = 2m$ or $2m + 1$.*

1. *If $n = 2m$ and $k = 2r + 1$, then*

$$B_{2m,2r+1}(-1) = 0, \quad B'_{2m,2r+1}(-1) = \frac{m!}{r!(m-r-1)!}. \quad (7.9)$$

Thus the zero is simple and the inverse at $y = 0$ is regular.

2. *In every other nontrivial case,*

$$B_{n,k}(-1) = \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, \quad B'_{n,k}(-1) = -\frac{D}{2} B_{n,k}(-1). \quad (7.10)$$

The inverse is again regular.

Proof. The value is the q -Lucas theorem at a primitive square root of unity. In the nonzero case D is even; differentiating reciprocity at $q = -1$ gives $2B'(-1) = -DB(-1)$. In the zero case, cancel the unique factor of $1 + q$ in the cyclotomic factorization and evaluate the remaining product, which gives (7.9). $\square$

Hence $q = -1$ itself creates no ramification for a nonconstant Gaussian polynomial: every level occurring there has nonzero first derivative. This contrasts sharply with the Pochhammer family, where special a values can force quadratic or cubic criticality.

7.6 Primitive roots of unity and cyclotomic valuation

Let ζ be primitive of order d and write $n = ad + b$, $k = rd + s$, with $0 \leq b, s < d$. The q -Lucas theorem states

$$B_{n,k}(\zeta) = \binom{a}{r} B_{b,s}(\zeta). \quad (7.11)$$

The exact exponent of the cyclotomic polynomial $\Phi_d(q)$ in $B_{n,k}$ is

$$e_d(n, k) = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{n-k}{d} \right\rfloor \in \{0, 1\}. \quad (7.12)$$

Thus whenever $B_{n,k}(\zeta) = 0$, the zero is simple. The local inverse at $y = 0$ is therefore analytic. At nonzero root-of-unity values, criticality is not forced by cyclotomy; it is detected by the derivative of the residual polynomial.

7.7 The complex branch locus

For fixed n, k , the finite critical values are the roots in y of

$$\text{Disc}_q(B_{n,k}(q) - y).$$

The radius of convergence of a local inverse series is bounded by the nearest critical value in the chosen y -plane. Since the coefficients are real and the polynomial is reciprocal, critical points occur in conjugate and reciprocal patterns. Determining their exact geometry is a finite algebraic problem for each rectangle, but no general closed classification appears to be known. This motivates [Theorem 13.2](#) below.

Chapter 8

Continuous Gaussian parameters

For $0 < q < 1$, define

$$\mathcal{B}_q(\alpha, \beta) = \frac{\Gamma_q(\alpha + 1)}{\Gamma_q(\beta + 1)\Gamma_q(\alpha - \beta + 1)}. \quad (8.1)$$

This interpolates the Gaussian polynomial at integer parameters and makes inversion in “ n ” and “ k ” an analytic problem.

8.1 Inversion in the upper parameter

Let $\psi_q = \partial_x \log \Gamma_q$. Then

$$\partial_\alpha \log \mathcal{B}_q = \psi_q(\alpha + 1) - \psi_q(\alpha - \beta + 1). \quad (8.2)$$

Because ψ_q is strictly increasing, this derivative is positive for $\beta > 0$ on the domain $\alpha > \beta - 1$.

Theorem 8.1 (Global continuous upper-parameter inverse). *Fix $0 < q < 1$ and $\beta > 0$. The function $\alpha \mapsto \mathcal{B}_q(\alpha, \beta)$ is strictly increasing on $(\beta - 1, \infty)$ from 0 to*

$$\frac{1}{(1 - q)^\beta \Gamma_q(\beta + 1)}. \quad (8.3)$$

It therefore has a unique real inverse for every level in this open range.

Proof. Monotonicity follows from (8.2). At the left endpoint the factor $\Gamma_q(\alpha - \beta + 1)$ has a pole. At infinity, the q -gamma quotient tends to $(1 - q)^{-\beta}$. $\square$

At a regular base point (α_0, y_0) , put $u = \log(y/y_0)$ and

$$c_m = \psi_q^{(m-1)}(\alpha_0 + 1) - \psi_q^{(m-1)}(\alpha_0 - \beta + 1).$$

Then the inverse series is exactly (2.1) with $x = \alpha - \alpha_0$.

8.2 Inversion in the lower parameter and the central branch point

Differentiation gives

$$\partial_\beta \log \mathcal{B}_q = \psi_q(\alpha - \beta + 1) - \psi_q(\beta + 1). \quad (8.4)$$

The function is symmetric under $\beta \mapsto \alpha - \beta$, so it reaches its unique maximum at $\beta = \alpha/2$. Put

$$A = \frac{\alpha}{2} + 1, \quad x = \beta - \frac{\alpha}{2}, \quad \mathcal{B}_0 = \mathcal{B}_q(\alpha, \alpha/2).$$

Taylor expansion gives

$$-\log \frac{\mathcal{B}_q(\alpha, \alpha/2 + x)}{\mathcal{B}_0} = \psi'_q(A)x^2 + \frac{\psi_q'''(A)}{12}x^4 + \mathcal{O}(x^6). \quad (8.5)$$

Hence, with $v = -\log(y/\mathcal{B}_0)$,

$$x = \pm \sqrt{\frac{v}{\psi'_q(A)}} \left(1 - \frac{\psi_q'''(A)}{24\psi'_q(A)^2}v + \mathcal{O}(v^2) \right). \quad (8.6)$$

This is a structural, not accidental, square-root inverse: the two lower parameters β and $\alpha - \beta$ are indistinguishable.

8.3 Continuous parameter inversion near $q = 1$

The q -gamma expansion in the next chapter has the form

$$\log \Gamma_{e^{-t}}(x) = \log \Gamma(x) + A_1(x)t + A_2(x)t^2 + \mathcal{O}(t^4).$$

Therefore

$$\log \mathcal{B}_{e^{-t}}(\alpha, \beta) = \log \binom{\alpha}{\beta} + \Delta A_1(\alpha, \beta)t + \Delta A_2(\alpha, \beta)t^2 + \mathcal{O}(t^4), \quad (8.7)$$

where

$$\Delta A_j = A_j(\alpha + 1) - A_j(\beta + 1) - A_j(\alpha - \beta + 1).$$

In particular,

$$\Delta A_1 = -\frac{\beta(\alpha - \beta)}{2}. \quad (8.8)$$

If α_0 solves the classical level equation $\binom{\alpha_0}{\beta} = y$ and

$$g_\alpha = \psi(\alpha_0 + 1) - \psi(\alpha_0 - \beta + 1) \neq 0,$$

then the corresponding q -inverse is

$$\alpha_q = \alpha_0 + \frac{\beta(\alpha_0 - \beta)}{2g_\alpha}t + \mathcal{O}(t^2). \quad (8.9)$$

Higher corrections follow mechanically from (2.3).

Chapter 9

Inverse theory for Jackson's q -gamma function

For $0 < q < 1$,

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}. \quad (9.1)$$

Its logarithmic derivative is

$$\psi_q(x) = -\log(1-q) + (\log q) \sum_{j \geq 0} \frac{q^{x+j}}{1-q^{x+j}}. \quad (9.2)$$

9.1 The two real x -inverse branches

Theorem 9.1 (Unique minimum of Γ_q). *For every $0 < q < 1$, ψ_q is strictly increasing from $-\infty$ to $-\log(1-q) > 0$ on $(0, \infty)$. Hence there is a unique $x_q^* > 0$ with $\psi_q(x_q^*) = 0$. The function Γ_q decreases on $(0, x_q^*)$, increases on (x_q^*, ∞) , and tends to $+\infty$ at both endpoints. Consequently:*

- *levels above $\Gamma_q(x_q^*)$ have exactly two positive real inverses;*
- *the minimum level has one double inverse;*
- *lower levels have no positive real inverse.*

Proof. Differentiating (9.2) gives

$$\psi'_q(x) = (\log q)^2 \sum_{j \geq 0} \frac{q^{x+j}}{(1-q^{x+j})^2} > 0.$$

The endpoint limits follow from the $j = 0$ term and from dominated convergence. The behavior of Γ_q then follows by integration and from (9.1). $\square$

Let $y_* = \Gamma_q(x_q^*)$. Since $\psi'_q(x_q^*) > 0$,

$$\log \frac{\Gamma_q(x)}{y_*} = \frac{\psi'_q(x_q^*)}{2} (x - x_q^*)^2 + \mathcal{O}((x - x_q^*)^3).$$

Thus the two branches coalesce as

$$x - x_q^* = \pm \sqrt{\frac{2 \log(y/y_*)}{\psi'_q(x_q^*)}} + \mathcal{O}(\log(y/y_*)). \quad (9.3)$$

9.2 The limit $q \rightarrow 1$ and inversion in x

Put $q = e^{-t}$. Moak's q -Stirling theory and the shifted-factorial asymptotics imply [9, 8]

$$\log \Gamma_{e^{-t}}(x) = \log \Gamma(x) + A_1(x)t + A_2(x)t^2 + \mathcal{O}(t^4), \quad (9.4)$$

with no cubic term and

$$A_1(x) = -\frac{(x-1)(x-2)}{4}, \quad A_2(x) = \frac{B_3(x)}{72} - \frac{x-1}{24}, \quad (9.5)$$

where $B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$.

Fix a regular classical inverse x_0 satisfying $\Gamma(x_0) = y$ and $\psi(x_0) \neq 0$. Write $x_q = x_0 + d_1t + d_2t^2 + \dots$. Then

$$d_1 = -\frac{A_1(x_0)}{\psi(x_0)}, \quad (9.6)$$

$$d_2 = -\frac{A_2(x_0) + A_1'(x_0)d_1 + \frac{1}{2}\psi'(x_0)d_1^2}{\psi(x_0)}. \quad (9.7)$$

The companion experiment for $x_0 = 3.4$ shows second-order errors decreasing as t^3 .

9.3 Inversion in the base near $q = 1$

Fix $x \notin \{1, 2\}$ and put $u = \log(y/\Gamma(x))$. Because $A_1(x) \neq 0$,

$$t = \frac{u}{A_1(x)} - \frac{A_2(x)}{A_1(x)^3}u^2 + \mathcal{O}(u^3), \quad q = e^{-t}. \quad (9.8)$$

At $x = 1$ and $x = 2$, $\Gamma_q(x) \equiv 1$, so no base inverse exists. These are genuine constant degeneracies, not missing terms in (9.8).

9.4 Inversion near $q = 0$

Expansion of (9.1) gives

$$\log \Gamma_q(x) = q^x + (x-2)q + \mathcal{O}\left(q^{\min(2, x+1, 2x)}\right). \quad (9.9)$$

The dominant term changes with x :

1. If $0 < x < 1$, then q^x dominates and

$$q \sim (\log y)^{1/x}. \quad (9.10)$$

This is a fractional-power inverse even when x is irrational.

2. If $x > 1$ and $x \neq 2$, then

$$q \sim \frac{\log y}{x-2}. \quad (9.11)$$

3. At $x = 1, 2$, the function is identically 1.

The full expansion lives in a Hahn series generated by the exponents 1 and x when x is nonintegral.

9.5 The standard $q > 1$ continuation and $q \rightarrow \infty$

Jackson's reciprocal-base continuation satisfies

$$\Gamma_q(x) = q^{(x-1)(x-2)/2} \Gamma_{1/q}(x), \quad q > 1. \quad (9.12)$$

Let $A = (x-1)(x-2)/2$. Since $\Gamma_{1/q}(x) \rightarrow 1$ as $q \rightarrow \infty$,

$$\Gamma_q(x) \sim q^A. \quad (9.13)$$

For $A \neq 0$, the base inverse has the leading Puiseux form

$$q \sim y^{1/A}, \quad (9.14)$$

with the branch selected by the reciprocal-base definition. Corrections are obtained by inserting the small- $1/q$ expansion (9.9) into (9.12). Again $x = 1, 2$ are constant exceptions.

Part V

Other q -analogs and transferable inverse schemes

Chapter 10

q -numbers, q -factorials, and q -exponentials

10.1 The q -number

For

$$[x]_q = \frac{1 - q^x}{1 - q},$$

inversion in x is elementary once logarithm branches are chosen:

$$x = \frac{\text{Log}(1 - y(1 - q))}{\text{Log } q}. \quad (10.1)$$

This exact formula is useful as a model: branch changes in Log create all complex inverse sheets, while the positive real branch is unique for $0 < q < 1$ and $y > 0$ in its range.

For base inversion, integer $x = n$ gives the polynomial $[n]_q = 1 + q + \cdots + q^{n-1}$. Near $q = 1$, set $q = e^t$ and $u = \log([x]_{e^t}/x)$. The Bernoulli expansion is

$$u = \frac{x-1}{2}t + \sum_{r \geq 1} \frac{B_{2r}(x^{2r} - 1)}{2r(2r)!} t^{2r}. \quad (10.2)$$

It is inverted by (2.1). At infinity, for integer $n \geq 2$ and $R = y^{1/(n-1)}$,

$$q = R - \frac{1}{n-1} + \mathcal{O}(R^{-1}). \quad (10.3)$$

For noninteger x , the equation $q^x - yq + (y-1) = 0$ defines a generalized algebraic-transcendental correspondence; Newton polygons in the exponents 1 and x replace ordinary polynomial analysis.

10.2 The q -factorial

Let

$$[n]_q! = \prod_{j=1}^n [j]_q, \quad N = \frac{n(n-1)}{2}.$$

At $q = 0$,

$$[n]_q! = 1 + (n-1)q + \frac{(n-2)(n+1)}{2}q^2 + \mathcal{O}(q^3).$$

Therefore, with $s = y - 1$,

$$q = \frac{s}{n-1} - \frac{(n-2)(n+1)}{2(n-1)^3} s^2 + \mathcal{O}(s^3). \quad (10.4)$$

At $q = e^t$ near 1,

$$\log \frac{[n]_{e^t}!}{n!} = \frac{N}{2} t + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} \sum_{j=1}^n (j^{2r} - 1) t^{2r}. \quad (10.5)$$

This is again a cumulant series with only one odd term. Reciprocity gives $[n]_q! = q^N [n]_{1/q}!$, and therefore

$$q = y^{1/N} - \frac{n-1}{N} + \mathcal{O}(y^{-1/N}). \quad (10.6)$$

10.3 Euler's two q -exponentials

Use

$$e_q(z) = \frac{1}{(z; q)_\infty}, \quad E_q(z) = (-z; q)_\infty.$$

For $0 < q < 1$, e_q maps $(-\infty, 1)$ strictly increasingly onto $(0, \infty)$, and E_q maps $(-1, \infty)$ strictly increasingly onto $(0, \infty)$. Thus both possess global positive-real inverses.

Put $w = y - 1$. Euler's series and reversion give

$$\begin{aligned} e_q^{-1}(1+w) &= (1-q)w - \frac{1-q}{1+q} w^2 \\ &\quad + \frac{(1-q)(2q^2+q+1)}{(1+q)^2(1+q+q^2)} w^3 + \mathcal{O}(w^4), \end{aligned} \quad (10.7)$$

$$\begin{aligned} E_q^{-1}(1+w) &= (1-q)w - \frac{q(1-q)}{1+q} w^2 \\ &\quad + \frac{q^2(1-q)(q^2+q+2)}{(1+q)^2(1+q+q^2)} w^3 + \mathcal{O}(w^4). \end{aligned} \quad (10.8)$$

Under the classical scaling $z = (1-q)x$, both direct functions tend to e^x , so their scaled inverses tend to $\log y$.

10.4 A normalized q -Lambert inverse

To obtain a deformation of the classical Lambert W with a regular $q \rightarrow 1$ limit, define $\mathcal{W}_q$ locally by

$$y = x e_q((1-q)x) \iff x = \mathcal{W}_q(y). \quad (10.9)$$

Since

$$x e_q((1-q)x) = x + x^2 + \frac{x^3}{1+q} + \mathcal{O}(x^4),$$

we obtain

$$\mathcal{W}_q(y) = y - y^2 + \frac{1+2q}{1+q} y^3 + \mathcal{O}(y^4). \quad (10.10)$$

As $q \rightarrow 1^-$, the coefficient tends to $3/2$ and $\mathcal{W}_q \rightarrow W$ locally. The critical points of the direct map satisfy

$$1 + (1-q)x \partial_z \log e_q(z) \big|_{z=(1-q)x} = 0,$$

and their images are the branch points of $\mathcal{W}_q$. Mapping their motion from $q = 0$ to $q = 1$ is a concrete continuation problem.

Chapter 11

Rogers–Szegő, q -Catalan, theta, and basic hypergeometric inverses

11.1 Rogers–Szegő polynomials

Let

$$H_n(z; q) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q z^k.$$

Inversion in z at the unit level is regular when $[n]_q \neq 0$:

$$z = \frac{y-1}{[n]_q} - \frac{\begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3} (y-1)^2 + \mathcal{O}((y-1)^3). \quad (11.1)$$

For inversion in q , the direct endpoint data are

$$\begin{aligned} H_n(z; 0) &= 1 + z + \cdots + z^n, \\ \partial_q H_n(z; 0) &= \sum_{k=1}^{n-1} z^k, \end{aligned} \quad (11.2)$$

$$\begin{aligned} H_n(z; 1) &= (1+z)^n, \\ \partial_q H_n(z; 1) &= \frac{n(n-1)}{2} z(1+z)^{n-2}. \end{aligned} \quad (11.3)$$

Thus the base inverse is locally linear at either endpoint unless the displayed slope vanishes. When it does vanish, the next nonzero q -derivative determines a Puiseux exponent. At $q = \infty$, the Newton polygon is controlled by the maximum of $k(n-k)$ among those k for which $z^k \neq 0$; a tie for odd n can change the leading coefficient and create additional exceptional levels.

11.2 MacMahon's q -Catalan polynomial

Define

$$\text{Cat}_n(q) = \frac{1}{[n+1]_q} \begin{bmatrix} 2n \\ n \end{bmatrix}_q.$$

For every $n \geq 2$,

$$\text{Cat}_n(q) = 1 + q^2 + \mathcal{O}(q^3). \quad (11.4)$$

The linear term cancels, so the inverse at $y = 1$ is necessarily ramified:

$$q = \pm\sqrt{y-1} + \mathcal{O}(y-1). \quad (11.5)$$

The positive sign is the canonical inverse on $q \geq 0$. At $q = 1$,

$$\text{Cat}_n(1) = \frac{1}{n+1} \binom{2n}{n}, \quad \frac{\text{Cat}'_n(1)}{\text{Cat}_n(1)} = \frac{n(n-1)}{2},$$

so the inverse is regular there for $n \geq 2$. This example shows that a ratio of two individually regular q -analogs can acquire a critical endpoint by coefficient cancellation.

11.3 Theta products

Let

$$\Theta(z; q) = (z; q)_\infty (q/z; q)_\infty, \quad 0 < |q| < 1. \quad (11.6)$$

Its zeros $z = q^m$, $m \in \mathbb{Z}$, are simple when q is not a root of unity, so the inverse in z is analytic at the zero level. The functional equations

$$\Theta(qz; q) = -z^{-1} \Theta(z; q), \quad \Theta(q/z; q) = \Theta(z; q)$$

organize the sheets. The involution $z \mapsto q/z$ has fixed points $z = \pm\sqrt{q}$; differentiation of the symmetry shows

$$\Theta_z(\pm\sqrt{q}; q) = 0.$$

Whenever the second derivative is nonzero, the inverse in z has a square-root branch at the corresponding theta value. This is the theta analogue of the central lower-parameter branch of the continuous Gaussian coefficient.

The logarithmic derivative

$$\partial_z \log \Theta(z; q) = - \sum_{j \geq 0} \frac{q^j}{1 - zq^j} + \frac{1}{z} \sum_{j \geq 0} \frac{q^{j+1}/z}{1 - q^{j+1}/z}$$

provides both local coefficients and a stable Newton iteration away from zeros and critical points.

11.4 A universal inverse template for ${}_r\phi_s$

Write a basic hypergeometric function in a fixed convention as

$$F(z) = {}_r\phi_s(\mathbf{a}; \mathbf{b}; q, z) = 1 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad c_1 \neq 0.$$

Then its local argument inverse is

$$z = \frac{y-1}{c_1} - \frac{c_2}{c_1^3} (y-1)^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4} \right) (y-1)^3 + \mathcal{O}((y-1)^4). \quad (11.7)$$

Because each c_m is a quotient of finite q -Pochhammer symbols, every parameter derivative can be written in finite Lambert sums. The same procedure therefore inverts in any numerator or denominator parameter at a regular point. At terminating parameters, coalescing zeros can force Puiseux series, and at $q \rightarrow 1$ the correct scaling should be taken before reversion to recover the corresponding ordinary hypergeometric inverse.

11.5 Products and ratios with many parameters

A broad class is

$$F(\mathbf{a}, q) = \prod_{\nu=1}^r (a_\nu; q)_{n_\nu}^{\varepsilon_\nu}, \quad \varepsilon_\nu \in \mathbb{Z}. \quad (11.8)$$

For a selected parameter a_μ ,

$$\partial_{a_\mu} \log F = -\varepsilon_\mu \sum_{j=0}^{n_\mu-1} \frac{q^j}{1 - a_\mu q^j}.$$

Thus all regular inverse coefficients follow from the universal calculus. The singular set is the union of factor zeros, poles, and the critical hypersurface on which the signed Lambert sums cancel. This description extends immediately to ratios defining orthogonal polynomials, q -beta functions, and many terminating basic hypergeometric summands.

Chapter 12

Series algorithms, continuation, and certified computation

12.1 A branch-aware workflow

A reliable inverse computation should separate symbolic branch analysis from numerical evaluation.

Algorithm 12.1 (Local and global inversion workflow). Given $F(x; \lambda) = y$:

1. Choose a base point (x_0, y_0, λ_0) and compute the first nonzero derivative of $F - y_0$ in x .
2. If the order is one, compute a Taylor inverse by Lagrange reversion or Newton iteration in a truncated series ring. If the order is $m > 1$, introduce $w = (y - y_0)^{1/m}$ and compute a Puiseux inverse.
3. Factor exact cyclotomic and Pochhammer zeros before numerical evaluation. Never evaluate a removable $0/0$ Gaussian product directly at a root of unity.
4. Near $q = 1$ or another unit-circle root, replace q by ζe^{-t} and invert the logarithmic asymptotic in t .
5. Continue the selected branch with a tangent predictor from (2.2) and a Newton corrector.
6. Monitor $|F_x|$. When it becomes small, switch to a local Puiseux coordinate or use pseudo-arclength continuation.
7. Certify the result by an interval derivative bound, interval Newton, or a complex Rouché test.

12.2 Formal reversion without expression swell

Direct substitution into a long symbolic product is expensive. Better methods are:

- work with logarithmic cumulants and Bell polynomials;
- use Newton iteration in the truncated ring $\mathbb{C}[[u]]/(u^{M+1})$, doubling the number of correct coefficients at each step;
- exploit coefficient stability: a finite product with $n > M$ has the same $q = 0$ inverse jet through order M as the infinite product;
- for Gaussian polynomials, obtain only the required rectangle partition numbers rather than expand the full polynomial;

- at infinity, work in $s = 1/q$ and use reciprocal symmetry.

If $g_0(u)$ is an initial inverse approximation, the truncated Newton update

$$g_{r+1}(u) = g_r(u) - \frac{F(g_r(u)) - u}{F'(g_r(u))} \pmod{u^{2^{r+1}}} \quad (12.1)$$

provides a fast general implementation.

12.3 Stable direct evaluation

Near $q = 1$, multiplying factors is badly conditioned. For $|a| < 1$, use the Lambert series (5.1); for finite products use compensated sums of logarithms. Near a root of unity, first group factors as in (5.11). For Gaussian coefficients at a root of unity, use cyclotomic factorization or the polynomial recurrence instead of the quotient of two vanishing Pochhammer products.

12.4 Numerical branch continuation

Suppose $F(q) = y$ and a branch is known at y_0 . The differential equation

$$\frac{dq}{dy} = \frac{1}{F'(q)}$$

provides a predictor. Close to a simple critical point, use $w = \sqrt{y - y_c}$ as the continuation parameter. For polynomial inverses, all branches can be tracked simultaneously by homotopy from a large level, where the Puiseux roots are explicitly separated. Matching at a moderate level then identifies the monodromy permutation induced by loops around critical values.

12.5 What the companion program checks

The distributed script `inverse_q_analogs_experiments.py` uses 80-digit arithmetic and exact SymPy polynomials. It checks:

- the finite/infinite stable $q = 0$ reversion and every low- n exception;
- singular inversion of $(a; q)_\infty$ at $q \rightarrow 1$;
- radial inversion at $q \rightarrow -1$;
- the fixed-level a -inverse at $q \rightarrow 1$;
- the Gaussian $q = 1$ inverse and exact $q = -1$ values and derivatives;
- the q -gamma inverse correction near $q = 1$;
- square-root ramification of the MacMahon q -Catalan polynomial;
- the exact x -inverse of the q -number.

The output is reproduced in [Chapter B](#).

Part VI

Frontier questions and synthesis

Chapter 13

Conjectures and research programs

13.1 Monodromy and generic Galois groups

Polynomial inversion naturally asks for the permutation group generated by continuation around the critical values.

Conjecture 13.1 (Generic Pochhammer monodromy). *Let $n \geq 3$ and $N = n(n-1)/2$. Over the rational function field $\mathbb{C}(a, y)$, the polynomial*

$$(a; q)_n - y$$

viewed as a polynomial in q has full symmetric geometric monodromy S_N , except on an explicitly classifiable finite union of degenerate parameter strata and low-degree cases.

Evidence should be sought by computing discriminants, branch cycles, and specializations with certified Galois groups. The large- y Puiseux branches from (4.8) give a natural base fiber for monodromy tracking.

Conjecture 13.2 (Generic Gaussian monodromy). *For rectangles outside an explicitly classifiable decomposable family, the polynomial map*

$$q \mapsto \begin{bmatrix} n \\ k \end{bmatrix}_q$$

has full symmetric monodromy S_D , $D = k(n-k)$, over a generic level. Reciprocity constrains the critical configuration but does not force an imprimitive inverse action for generic y .

This formulation intentionally allows exceptions. Composition identities, small rectangles, and accidental critical-value collisions must be removed before a full-symmetric statement can be true.

13.2 All-orders root-of-unity inversion

Conjecture 13.3 (Cyclotomic polylogarithm closure). *For primitive ζ of order d , the complete radial expansion of $\text{Log}(a; \zeta e^{-t})_\infty$ has coefficients in the differential algebra generated by*

$$\text{Li}_m(a\zeta^r), \quad \text{Log}(1 - a\zeta^r), \quad 0 \leq r < d,$$

with rational functions of a and cyclotomic constants. After inversion, the base series belongs to the corresponding algebra in odd inverse powers of the large logarithmic coordinate, up to exponentially small sectors.

The first part should follow systematically from Euler–Maclaurin expansion of each factor in (5.11). The difficult part is a uniform statement across branch cuts and Stokes directions.

Problem 13.4 (Uniform $a \asymp t^{1/d}$ expansion). *Construct a two-parameter uniform expansion for $(a; \zeta e^{-t})_\infty$ when $a = t^{1/d}A$ and A ranges over compact complex sets. Determine the complete inverse series $A = A(y, t)$ and its Stokes transitions. The formulas (5.10) and (5.17) are the first two cases.*

13.3 Resurgence and exponentially small inverse corrections

The McIntosh series is generally divergent at sufficiently high order. Its least term predicts an exponentially small error associated with modular or saddle contributions. Inverting a divergent expansion is not accomplished by formal reversion alone: exponentially small changes in $\log(a; q)_\infty$ can be amplified by the inverse condition number.

Problem 13.5 (Resurgent inverse q -Pochhammer). *Determine the Borel singularities and Stokes constants of the $q \rightarrow 1$ and $q \rightarrow \zeta$ expansions, then derive the induced transseries for the inverse base $q(y)$. Quantify how the inverse Stokes jump depends on $\text{Li}_2(a^d)$ and on proximity to a critical level.*

A promising method is to combine modular transformations of eta/theta specializations with exact WKB-style inversion of the logarithmic coordinate.

13.4 Critical values, zero attractors, and inverse conditioning

For fixed (n, k) , the critical points of a Gaussian polynomial and their images can be computed exactly, but their large-rectangle distribution is largely unexplored.

Problem 13.6 (Limit distribution of Gaussian critical values). *Let $k/n \rightarrow \rho \in (0, 1)$. After the natural logarithmic scaling in q and in the level, determine the limiting distribution of critical points and critical values of $[n]_q^k$. Relate it to the limit shape of partitions in a rectangle and to saddle points of the associated free energy.*

Such a result would determine asymptotic radii of convergence for inverse series at $q = 0$ and $q = 1$ and would identify the best domains for rational or conformal acceleration.

13.5 Coefficient structure of inverse series

The first inverse coefficients often display positivity or alternating signs, but exact computations show that naive global sign conjectures fail even for small rectangles. The right question is more refined.

Problem 13.7 (Combinatorics of reversion coefficients). *Find signed combinatorial models for*

$$[s^m]Q_{n,k}(1+s) \quad \text{and} \quad [x^m]Q_n((1-a) + (a^2 - a)x; a)$$

in terms of decorated partitions, plane trees, or cumulant assemblies. Classify the first sign changes and determine whether natural rescalings yield log-concavity, total positivity, or finite-order sign regularity.

Lagrange inversion already expresses the coefficients as weighted tree sums; the missing step is a weight-preserving interpretation adapted to rectangle partitions or Pochhammer subset sums.

13.6 Multivariate and elliptic extensions

A basic hypergeometric identity supplies an algebraic or analytic hypersurface in all of its parameters. Multivariate inversion can therefore reveal relations invisible in one-variable specializations.

Problem 13.8 (Inverse basic hypergeometric varieties). *Develop a branch atlas for simultaneous inversion in $(z, q, a_1, \dots)$ of terminating ${}_r\phi_s$ functions. Identify discriminant components caused by termination, balancing, and root-of-unity resonance. Extend the method to elliptic shifted factorials and elliptic gamma functions, where two nomes produce interacting unit-circle singularities.*

The proper local tool is the multivariate Weierstrass preparation theorem; the proper global tool is numerical algebraic geometry combined with exact special-function identities.

13.7 Connections to geometric-comb interpolation and atomic functions

The source monograph emphasizes Newton and Lagrange interpolation on geometric nodes. The basis factors of such interpolation are shifted products

$$\prod_{j=0}^{m-1} (x - q^j),$$

which are elementary transforms of q -Pochhammer symbols. Their inverse level sets determine where the basis is well conditioned and where several geometric nodes create ramification. The complex-order inverse (6.5) also gives an “effective depth” of a geometric tail: it answers how many dyadic or q -adic scales are needed to achieve a given product level.

For the Rvachev up-function and the Fabius function, Fourier transforms and functional equations involve products over dyadic scales. Although the factors are sinc functions rather than literal shifted factorials, the same architecture applies: logarithmic cumulants for regular inversion, Lambert W -type coordinates for exponentially flat limits, and Puiseux branches when symmetry forces criticality. A concrete research direction is to transfer the certified branch-continuation machinery of this report to inverse level sets of infinite sinc products and then compare their effective-depth variable with α in (6.1).

13.8 Formal verification

A realistic Lean development can be modular.

1. Formal power-series reversion and Lagrange–Bürmann coefficients.
2. A local Puiseux theorem for a factored holomorphic germ, initially in a formal setting.
3. Finite q -Pochhammer derivative and zero formulas such as (3.3).
4. Positivity and monotonicity proofs for the global real branches.
5. Cyclotomic factorization and the q -Lucas evaluation.
6. The exact $q = -1$ formulas in Theorems 4.3 and 7.2.
7. An asymptotic algebra supporting substitution and inversion of Poincaré series, culminating in (5.8).

The finite algebraic layers are suitable for immediate formalization. The uniform analytic and resurgent layers require a larger asymptotic library.

Chapter 14

Conclusions

The inverse theory of q -analogs is not a collection of isolated numerical root-finding problems. It has a coherent architecture:

- products suggest logarithmic cumulants;
- ordinary points give Taylor reversion;
- zero multiplicities and symmetry give Puiseux exponents;
- reciprocal polynomials give expansions at $\pm\infty$;
- cyclotomic factorization controls roots of unity;
- dilogarithmic singularities at the unit circle invert in a large logarithmic coordinate;
- continuous parameters are governed by q -digamma monotonicity;
- global real branches can often be isolated before any complex analysis.

The most novel synthesis is the root-of-unity law (5.12): it unifies $q \rightarrow 1$ and radial $q \rightarrow -1$ and predicts the fixed-level inverse exponent $1/d$ at every primitive d -th root. The finite theory mirrors it exactly through residue-block factorization. Together with the full branch atlas at infinity and the continuous-parameter analysis, this gives a practical foundation for a much larger program of inverse basic hypergeometric and geometric-scale functions.

Appendix A

Compact coefficient atlas

A.1 Universal regular inverse

If $u = c_1x + c_2x^2/2! + c_3x^3/3! + c_4x^4/4! + \dots$, then

$$x = \frac{u}{c_1} - \frac{c_2u^2}{2c_1^3} + \frac{(3c_2^2 - c_1c_3)u^3}{6c_1^5} + \frac{(-c_1^2c_4 + 10c_1c_2c_3 - 15c_2^3)u^4}{24c_1^7} + \dots$$

A.2 Finite and infinite Pochhammer quick reference

Regime	Inverse expansion
$a \rightarrow 0$, finite	$a = z/E_1 + E_2z^2/E_1^3 + (2E_2^2/E_1^5 - E_3/E_1^4)z^3 + \dots$, $z = 1 - y$
$q \rightarrow 0$, finite/infinite	$q = x - x^2 + (a+1)x^3 - (4a+1)x^4 + (3a^2 + 11a + 1)x^5 + \dots$
$q \rightarrow 1$, finite	$u = \sum_{m \geq 1} -S_m(n) \text{Li}_{1-m}(a)t^m/m!$, followed by universal reversion
$q \rightarrow \infty$, finite	$s = xH(s)^{1/N}$, $[x^m]s = m^{-1}[u^{m-1}]H(u)^{m/N}$, $q = 1/s$
$a \rightarrow 0$, infinite	$a = (1-q)z + q(1-q)z^2/(1+q) + \dots$
$q \rightarrow 1$, infinite fixed a	$t = A/\Lambda + A^2C/\Lambda^3 + (2A^3C^2 - A^4E)/\Lambda^5 + \dots$
$q \rightarrow 1$, infinite fixed	$a = Lt - L(L+2)t^2/4 + L(L^2 + 9L + 12)t^3/72 + \dots$
$y = e^{-L}$	
$q \rightarrow \zeta$, order d	$\log(a; q)_\infty = -\text{Li}_2(a^d)/(d^2t) + C_{d,\zeta}(a) + \mathcal{O}(t)$
a at fixed level near ζ	$a \sim \omega(d^2Lt)^{1/d}$
complex order $\alpha \rightarrow \infty$	$\alpha = (\text{Log}((1-q)w/a) - w/(1+q) + \dots)/\text{Log } q$

A.3 Gaussian quick reference

Regime	Inverse expansion
$q \rightarrow 0$	$q = s - p_2s^2 + (2p_2^2 - p_3)s^3 + \dots$, $s = y - 1$
$q \rightarrow 1$	$t = 2u/D - (n+1)u^2/(3D^2) + (n+1)^2u^3/(9D^3) + \dots$
$q \rightarrow \infty$	$q = y^{1/D} - D^{-1} + ((D-1)/(2D^2) - p_2/D)y^{-1/D} + \dots$
$q = -1$, zero case	$B'_{2m,2r+1}(-1) = m!/[r!(m-r-1)!]$

Regime	Inverse expansion
$q = -1$, nonzero case	$B'(-1) = -(D/2)B(-1)$
continuous β at center	$\beta - \alpha/2 = \pm \sqrt{v/\psi'_q(\alpha/2 + 1)}(1 + \mathcal{O}(v))$

Appendix B

Numerical and symbolic validation

The following output is generated by the companion script. The calculations use 80 decimal digits; symbolic checks are exact.

```
NUMERICAL AND SYMBOLIC VALIDATION
=====
mpmath working precision: 80 decimal digits

EXPERIMENT 1: symbolic q=0 reversion for (a;q)_n
n=2: q(x)=x
n=3: q(x)=3*a**2*x**5 + 21*a*x**5 - 5*a*x**4 + a*x**3 + 14*x**5 - 5*x**4 + 2*x**3 - x**2 + x
n=4: q(x)=3*a**2*x**5 + 10*a*x**5 - 4*a*x**4 + a*x**3 - 4*x**5 + x**3 - x**2 + x
n=5: q(x)=3*a**2*x**5 + 11*a*x**5 - 4*a*x**4 + a*x**3 + 2*x**5 - x**4 + x**3 - x**2 + x
n=6: q(x)=3*a**2*x**5 + 11*a*x**5 - 4*a*x**4 + a*x**3 + x**5 - x**4 + x**3 - x**2 + x
n=8: q(x)=3*a**2*x**5 + 11*a*x**5 - 4*a*x**4 + a*x**3 + x**5 - x**4 + x**3 - x**2 + x
Stable formula verified exactly for n>=6 through x^5.

EXPERIMENT 2: numerical q=0 inverse for finite q-Pochhammer
a=0.37, n=8; x=(y-(1-a))/(a^2-a)
q=0.02 err(0(x^3))=4.114876e-7 err(0(x^5))=9.334197e-10
q=0.01 err(0(x^3))=2.525137e-8 err(0(x^5))=1.409753e-11
q=5.0e-3 err(0(x^3))=1.563988e-9 err(0(x^5))=2.165947e-13

EXPERIMENT 3: singular q->1 inversion for (a;q)_infinity
a=0.30; t=-log q is reconstructed from y
t=0.1 err1=1.093415e-4 err3=1.908121e-7 err5=4.322132e-10
t=0.07 err1=3.75335e-5 err3=3.212219e-8 err5=3.56991e-11
t=0.05 err1=1.368343e-5 err3=5.977214e-9 err5=3.391259e-12
t=0.035 err1=4.694337e-6 err3=1.005001e-9 err5=2.794899e-13

EXPERIMENT 4: radial q->-1 inversion for (a;q)_infinity
q=-e^{-t}, a=0.30; t is reconstructed from y
t=0.1 err1=3.782447e-3 err3=2.755178e-4
t=0.07 err1=1.32388e-3 err3=4.867093e-5
t=0.05 err1=4.871469e-4 err3=9.267835e-6
t=0.035 err1=1.679576e-4 err3=1.577654e-6

EXPERIMENT 5: a-inverse of (a;e^{-t})_infinity at fixed y=e^{-1}
t=0.1 exact a=0.0927986164814 |e2|=2.986165e-4 |e3|=6.939074e-6 |e4|=1.789814e-7
t=0.07 exact a=0.0664281266999 |e2|=1.031267e-4 |e3|=1.678856e-6 |e4|=3.01895e-8
t=0.05 exact a=0.0481627551921 |e2|=3.775519e-5 |e3|=4.392523e-7 |e4|=5.626148e-9
t=0.035 exact a=0.0340942448263 |e2|=1.299483e-5 |e3|=1.058682e-7 |e4|=9.471694e-10

EXPERIMENT 6: q=1 inverse of [8 choose 3]_q
u=log(y/binom(8,3)); compare two- and three-term t-series
t=0.08 err1=4.785489e-3 err2=6.05945e-4 err3=7.972814e-5
t=0.04 err1=1.199089e-3 err2=7.393444e-5 err3=4.736688e-6
t=-0.04 err1=1.199089e-3 err2=6.995627e-5 err3=4.239191e-6
```

EXPERIMENT 7: exact Gaussian coefficients at $q=-1$

(n,k)=(6,3): value=0, derivative=6
(n,k)=(8,3): value=0, derivative=12
(n,k)=(7,3): value=3, derivative=-18
(n,k)=(8,4): value=6, derivative=-48
(n,k)=(9,4): value=6, derivative=-60
All displayed value/derivative formulas verified exactly.

EXPERIMENT 8: inverse $\Gamma_q(x)=\Gamma(3.4)$ as $q \rightarrow 1$

predicted $x_q=3.4+(0.785363930588)t+(0.385294212679)t^2+\dots$
t=0.08 exact x=3.465387295467 |e1|=2.558181e-3 |e2|=9.229806e-5
t=0.05 exact x=3.440253682381 |e1|=9.854859e-4 |e2|=2.225032e-5
t=0.03 exact x=3.423912448689 |e1|=3.515308e-4 |e2|=4.76598e-6
t=0.025 exact x=3.419877659489 |e1|=2.435612e-4 |e2|=2.752342e-6

EXPERIMENT 9: square-root ramification for MacMahon q -Catalan

n=5; compare q with $\sqrt{\text{Cat}_5(q)-1}$
q=0.04 $\sqrt{y-1}=0.0408575261454$ difference=8.575261e-4
q=0.02 $\sqrt{y-1}=0.0202070926821$ difference=2.070927e-4
q=0.01 $\sqrt{y-1}=0.0100508807087$ difference=5.088071e-5

EXPERIMENT 10: exact x -inverse of $[x]_q$

x=0.7 recovered=0.7 error=1.05422e-81
x=2.5 recovered=2.5 error=0.0
x=7.2 recovered=7.2 error=8.433758e-81

Appendix C

Companion-code notes

The archive includes `inverse_q_analogs_experiments.py`. The script is deliberately independent of the LaTeX source. Its main implementation choices are:

- the Lambert series $-\sum_{m \geq 1} a^m / [m(1 - q^m)]$ for stable infinite-product logarithms;
- exact Gaussian polynomials from the Pascal recurrence, avoiding removable root-of-unity singularities;
- direct coefficient-by-coefficient symbolic reversion at $q = 0$;
- high-precision root solving only after an asymptotic formula supplies a branch-specific initial value;
- assertions for every exact $q = -1$ value and derivative formula.

Run it from the report directory with

```
python inverse_q_analogs_experiments.py
```

It overwrites `numerical_results.txt` with a reproducible report.

Appendix D

Bibliographic notes

The direct q -product and Gaussian theory is classical; standard references include Gasper–Rahman [3], Andrews–Askey–Roy [4], Ismail [5], and the NIST Digital Library of Mathematical Functions [7]. McIntosh [8] supplies the shifted-factorial asymptotic central to the singular inverse at $q = 1$; Moak [9] develops the q -Stirling framework. Lagrange inversion, analytic combinatorics, and branch singularities are treated in [10, 11, 12]. The repository sources [1, 2] provide the immediate context and notation for this report.

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